T1 误差 $\frac{1}{2}\cdot 10^{-2}=0.005$

检查：$f(1)=-1$，$f(1.5)=3.375-2.5=0.875>0$

$\therefore f(1)\cdot f(1.5)<0$. 由介值定理，内含至少一根.

n	a_n	b_n	x_n	$f(x_n)$	Sign.
1	1.0	1.5	1.25	−0.2969	−
2	1.25	1.5	1.375	0.2246	+
3	1.25	1.375	1.3125	−0.0515	−
4	1.3125	1.375	1.34375	0.0826	+
5	1.3125	1.34375	1.328125	0.0146	+
6	1.3125	1.328125	1.3203125	−0.0187	−

$\because |1.3125-1.328125|/2>\frac{1}{2}\cdot 0.01$

$\therefore$ 取 $x=1.32$ 即可.

T2. $f(x)=0 \triangleq x=\varphi(x)$. 其中迭代函数形如

$x=\varphi(x)-\lambda f(x)$

$\varphi'(x)=1-\lambda f'(x)$，又 $\because \forall x, \exists f'(x) \wedge 0<m\le f'(x)\le M$

$\therefore$ 各项乘 $(-\lambda)$，有.

$-\lambda m\ge -\lambda f'(x)\ge -\lambda M$

则 $1-\lambda m\ge 1-\lambda f'(x)\ge 1-\lambda M$

即 $1-\lambda m\ge \varphi'(x)\ge 1-\lambda M$

$\because \lambda>0, m>0.$ $\therefore 1-\lambda m<1.$

又由 $0<\lambda<\frac{2}{M}$，则 $M\lambda<2$. 即 $1-\lambda M>-1$.

$\therefore -1<\varphi'(x)<1$

$\Rightarrow |\varphi'(x)|<1$

由压缩映射，$\{x_n\}$ 均收敛于 $f(x)=0$ 的根.

T3. Newton 迭代 $x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$

本题 $f(x)=x^3-3x-1$

$f'(x)=3x^2-3$，取 $x_0=1.9$

$x=1.9$，$f(1.9)=1.9^3-3(1.9)-1=0.159$

$f'(1.9)=3(1.9)^2-3=7.83$

$x'=1.9-\frac{0.159}{7.83}\approx 1.87969$

$x=1.87969$，$f(x)=0.0025$.

$f'(x)=7.5997$

$x'=x-\frac{0.0025}{7.5997}\approx 1.87936$

$x=1.87936$，$x-x_{pre}<\frac{1}{2}\cdot 10^{-3}$

$\therefore$ 取 $x=1.879$